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Emissions of HC, CO, NO_x, CO_2, and SO_2 from civil aviation in China in 2010

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ABSTRACT

Civil aviation in China has developed rapidly in recent years, and the effects of civil aviation emissions on the atmospheric environment should not be neglected. The establishment of emission inventories of atmospheric pollutants from civil aviation contributes to related policy formation and pollution control. According to the 2010's China flight schedules, aircraft/engine combination information and revised emission indices from the International Civil Aviation Organization emission data bank based on meteorological data, the fuel consumption and HC, CO, NO_x, CO_2, SO_2 emissions from domestic flights of civil aviation in China (excluding Taiwan Province) in 2010 are estimated in this paper. The results show that fuel consumption in 2010 on domestic flights in China is 12.12 million tons (metric tons), HC, CO, NO_x, CO_2 and SO_2 emissions are 4600 tons, 39,700 tons, 154,100 tons, 38.21 million tons and 9700 tons, respectively. The fuel consumption and pollutant emissions of China Southern Airline are responsible for the largest national proportion of each, accounting for 27% and 25–28%, respectively.

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1. Introduction

With the rapid growth of China's national economy in recent years, civil aviation industry has recorded rapid development. In 2010, China's civil aviation industry achieved a transport turnover (including passenger and cargo turnover) volume of 53.845 billion ton·km, an increase of 11.138 billion ton·km over the previous year. Therein, the passenger turnover volume was 35.955 billion ton·km, 5.871 billion ton·km more than that of the previous year (CAAC, 2011). Despite of civil aviation industry's great contribution to national economy (1.03% of China's GDP in 2010) (CAAC, 2011), it emits various pollutants such as hydrocarbons (HC), carbon monoxide (CO), nitrogen oxides (NO_x), carbon dioxide (CO_2) and sulfur dioxide (SO_2) into the atmosphere. As civil aircrafts mainly operate in the upper troposphere and lower stratosphere, these pollutants not only degrade air quality around the airports, but also deplete the ozone layer and harm the atmospheric environment (Colvile et al., 2001). Controlling aviation emissions is therefore of great significance for improvement of environmental quality and energy economy. The European Union (EU) required the aviation industry to undertake the obligation of reducing emissions by legislation in 2008. Several airlines in China will be included in the EU carbon emission trading system by 2012 (Ma and Feng, 2010), meaning that the emission of pollutants will impact the operation and performance of the aviation industry.

Establishment of pollutant emission inventories is one of the basic and effective moves to assess pollution, simulate environment, and develop related standards and pollution control strategies. A complete flight mission can be divided into two parts: the LTO (Landing and Take-off) cycle and the cruise process. Thus the assessment of aircraft emissions generally includes two parts: the assessment of airport ground-level pollutant emissions and evaluation of emissions during the cruise process.

The standard LTO cycle defined by the International Civil Aviation Organization (ICAO) consists of four operation processes: taxiing, take-off, climb, and approach. The emissions of LTO cycle include emissions from the land surface to the top of the atmospheric boundary layer of 915 m high (Kesign, 2006). In order to estimate the LTO cycle fuel consumption and pollutant emissions, an engine emission data bank was established by ICAO according to validation data from engine manufacturers, and was utilized to calculate the LTO cycle emissions of HC, CO and NO_x (Kalivoda and Feller, 1995; Kesign, 2006; Pejovic et al., 2008; Xia et al., 2008).

As most of the pollutants from aircrafts are emitted during the cruise process (Wilkerson et al., 2010), evaluation of emissions from this process is very important for assessing the impact of aviation to regional or worldwide atmospheric environment. NASA estimated the emissions of HC, CO and NO_x during the cruise process of subsonic jet aircrafts through the SASS (Subsonic Assessment)

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Program (Friedl et al., 1997). As for estimation of high altitude emissions of NO_x, the EU established the ANCAT (Abatement of Nuisances Caused by Air Transport) three-dimensional (longitude, latitude, and altitude) database (Gardner et al., 1997) and upgraded it to a four-dimensional one by adding the time axis in its latest version (Gardner et al., 1998).

Considering that pollutant emissions from aviation in China are increasing significantly and account for larger proportion in global amounts in recent years (Wilkerson et al., 2010), establishment of detailed emission inventories of various pollutants during both the LTO and cruise processes for China's aviation is of great significance. To our knowledge, however, only Ma and Zhou (2000) developed a three-dimensional inventory of aircraft NO_x emissions over China for a calendar year of 1997–1998. Although several global aviation emission inventories have been established (Eyers et al., 2004; Kim et al., 2005), detailed inventories for China are not publicly available to date.

Based on the 2010's China flight schedule, aircraft/engine combination information, and revised emission indices from the ICAO emission data bank with meteorological data, this paper estimates the fuel consumption and emissions of HC, CO, NO_x, CO_2, SO_2 from domestic flights of civil aviation in China in 2010.

2. Inventory methodology and data sources

2.1. Range of study

This paper assesses the HC, CO, NO_x, CO_2, SO_2 emissions from 29 commercial airlines of China (excluding Taiwan Province) during their operational period of domestic flights in 2010.

2.2. Data sources

1) Schedule database

The schedules used in this paper are from the planning schedules of the 29 airlines. The airline name, flight number, airplane type, flight date, starting place and time, terminal place and time, are given for each flight. The flights have been arranged as a form of city-pair so that some miscalculations such as flight double counting can be avoided.

2) Aircraft/engine combinations

Aircraft/engine combinations are picked from official websites of aircraft manufacturers. For the same airline, aircrafts of one specific type may be equipped with different types of engines. But the detailed aircraft/engine combination information is difficult to approach. To simplify analysis, we have chosen the most typical combinations according to combination information from the aircraft manufacturers and the ICAO emission data bank (ICAO, 2010). The aircraft/engine combinations used for calculation are listed in Table 1.

3) Engine emission indices

The ICAO emission data bank includes HC, NO_x and CO emission indices of different types of jet engines measured under the ISA (International Standard Atmosphere) condition at sea level. In addition, the CO_2 and SO_2 emission indices used in this paper are 3155 g kg^{-1}-Fuel and 0.8 g kg^{-1}-Fuel, respectively (Baughcum et al., 1996).

Table 1
Typical aircraft/engine combinations used for calculation.

Aircraft type	Engine type	Aircraft type	Engine type
A300	CF6-50C2	B737-800	CFM56-7B27
A300-600(R)	CF6-80C2	B757	RB211-535E4
A319	CFM56-5B7/P	B767	CF6-80C2
A320	CFM56-5B6	B767-300	PW4060
A321	CFM56-5B2/2P	B777	GE90-115B
A323	IAE V2530-A5	B777-200	GE90-115B
A330	CF6-80E1	CRJ	CF34-3A1
A330-300	CF6-80E1	CRJ200	CF34-3B1
A340	CFM56-5C4	CRJ700	CF34-8C5B1
A340-300	CFM56-5C4	E190	CF34-10E
B737	CFM56-3C-1	MD-90	IAE V2528-D5
B737-300/400	CFM56-3C-1	RJ145	AE 3007A
B737-700	CFM56-7B26		

2.3. Assumptions

To simplify analysis, we have made the following assumptions:

1) An airline between two cities is seen as a Great Circle route, i.e., each city is seen as a point, and aircrafts fly in a straight line between two points.
2) Aircrafts perform as the designed performance. The cruise speed is valued under the designed conditions. The cruise altitude is set to 11 km in this paper, as major aircrafts cruise at altitude between 9 and 13 km (Daggett et al., 1999), and pollutants emitted at around 11 km account for the largest percentage of total emissions (Ma and Zhou, 2000).
3) The load factor of all flights is assumed to be 70%. Aircrafts are loaded with no cargoes except passengers' luggage.
4) The fuel carried by each flight is exactly sufficient.
5) At the same cruise altitude and speed, the reduction of fuel flow caused by lightening of the aircraft weight along with fuel consumption is not taken into account.
6) The stages forming a flight, their time allocation and the requirements of engine thrust are as follows: Take-off process demands 100% engine thrust and takes 40 s; Climb process demands 85% engine thrust and takes 20 min; Cruise process demands 70% of engine thrust, of which time consumed is calculated with the distance and cruise speed of the flight; Approach process demands 30% engine thrust and takes 20 min; Taxiing and idle process demands 7% engine thrust and takes 26 min.
7) All flights fly in the standard atmosphere.
8) All flights fly as planned.
9) Wind and changes of temperature and humidity in a whole day are neglected.
10) The emissions are added up simply, neglecting the chemical reaction, drift and diffusion of the pollutants in the atmosphere.

2.4. Calculation method

The emissions of HC, CO, and NO_x are calculated by using the Boeing Fuel Flow Method 2 (BFFM2) in this study through the following three steps: calculation of fuel flow, revision of emission indices and calculation of emissions. BFFM2 has been widely used to evaluate the emissions of HC, CO, and NO_x, since it is an open and publicly available method (Baughcum et al., 1996; Daggett et al., 1999; Kim et al., 2005). As the emission indices of CO_2 and SO_2 are based strictly on fuel composition, the emissions of CO_2 and SO_2 are directly calculated by multiplying the fuel consumption with the corresponding emission index.

2.4.1. Calculation of fuel flow

The fuel flow we are dealing with consists of two parts: the LTO fuel flow and the cruise fuel flow. The LTO fuel flow of engines has been given in the ICAO emission data bank. What needs to be calculated is the cruise fuel flow. For the fuel flow given by the ICAO emission data bank, installation effects are neglected. Therefore, before calculating the actual fuel flow, we need to revise the data from the database so that the results can conform to the actual fuel flow better. The correction coefficients of the fuel flow for take-off, climb, approach and taxiing processes are 1.010, 1.013, 1.020 and 1.100 respectively. There is a linear relationship between the corrected fuel flow and engine thrust level. Placing the data of corrected fuel flow relating to its thrust level into Cartesian coordinates, the fuel flow under the cruise thrust setting (W_{ff}) can be calculated with linear regression method. However, this fuel flow is not the one of aircrafts flying at high altitude. Because tests by ICAO were taken on the ground, revision of the altitude has to be made in order to estimate the fuel flow of aircrafts flying at high altitude.

The cruise fuel flow can be calculated in accordance with the following equations (Baughcum et al., 1996):

$$W_f = \frac{W_{ff}}{\theta_{amb}^{3.8}\exp(0.2M^2)}\delta_{amb} \quad (1)$$

$$\theta_{amb} = (T_{amb} + 273.15)/288.15 \quad (2)$$

$$\delta_{amb} = P_{amb}/101.325 \quad (3)$$

where W_f and W_{ff} are fuel flow at flight altitude and sea level condition respectively (kg s^{-1}), θ_{amb} is the ratio of inlet temperature over sea level temperature, δ_{amb} is the ratio of inlet pressure over sea level pressure, P_{amb} and T_{amb} are ambient pressure (kPa) and ambient temperature (°C) respectively, M is the cruise Mach number of aircrafts.

Fuel flow multiplied by flight duration is fuel consumption.

2.4.2. Revision of emission indices

As the emission indices of HC, CO, and NO_x given in the ICAO emission data bank are indices under ISA condition at sea level, it is required to revise the emission indices with existing data to obtain indices under actual cruise conditions.

To facilitate distinction, emission indices measured by ICAO at sea level are called REI (Reference Emission Indices). Plot the REIs against the corrected W_f on log–log coordinate system, and we can find that the HC and CO are bi-linear least square fitted curves. Extrapolating both curves to the point of intersection gives the bi-linear relationship. Some engine emissions data sets do not fit this scheme well and must be manually manipulated. The NO_x curve is a simple point-to-point linear fit. We can get the REIs of each pollutant from the log–log coordinate system corresponding to the W_f figured out in Section 2.4.1.

The REIs obtained by the above method are the indices under ISA condition. We need to correct the REIs as follows:

$$EI(HC) = REI(HC)\theta_{amb}^{3.3}\Big/\delta_{amb}^{1.02} \quad (4)$$

$$EI(CO) = REI(CO)\theta_{amb}^{3.3}\Big/\delta_{amb}^{1.02} \quad (5)$$

$$EI(NO_x) = REI(NO_x)\exp(H)\left(\theta_{amb}^{3.3}\Big/\delta_{amb}^{3.02}\right)^{-0.5} \quad (6)$$

The variable H appearing in Eq. (6) can be calculated by the following equations:

$$H = -19.0 \times (\omega - 0.0063) \quad (7)$$

$$\omega = \frac{0.62198(\varphi)P_v}{P_{amb} - (\varphi)P_v} \quad (8)$$

$$P_v = (0.014504) \times 10^{\beta} \quad (9)$$

$$\begin{aligned}\beta = {} & 7.90298\left(1 - \frac{373.16}{T_{amb}+273.16}\right) + 3.00571 \\ & + (5.02808)\lg\left(\frac{373.16}{T_{amb}+273.16}\right) \\ & + \left(1.3816 \times 10^{-7}\right)\left[1 - 10^{11.344\left(1-\frac{T_{amb}+273.16}{373.16}\right)}\right] \\ & + \left(8.1328 \times 10^{-3}\right)\left[10^{3.49149\left(1-\frac{373.16}{T_{amb}+273.16}\right)} - 1\right]\end{aligned} \quad (10)$$

where ω and φ are specific humidity and relative humidity at the cruise altitude respectively, P_{amb} and P_v are ambient pressure and saturated vapor pressure at the cruise altitude (kPa).

In this paper, we revise the emission indices of HC, CO, and NO_x from the ICAO emission data bank with meteorological parameters during 1971–2000 in China.

2.4.3. Calculation of emissions

Emissions of each pollutant for a particular flight can be calculated as follows:

$$\begin{aligned}S(HC, CO, NO_x, CO_2, SO_2) = {} & N \times \sum_i^n EI_i(HC, CO, NO_x, CO_2, SO_2) \\ & \times W_{f_i} \times T_i \times 10^{-3}\end{aligned} \quad (11)$$

where S is the pollutant emission for a particular flight (kg), N is the number of engines, T_i is the duration of flight process i (min).

Monthly emissions of a particular flight number can be estimated by multiplying the above emissions by flight frequency in one month. By adding up the emissions of each month, we can get the annual emissions. By adding up the annual emissions of each flight number, we can figure out the total emissions produced by aviation industry in a year.

3. Results and discussion

3.1. Fuel consumption

Following the method described in Section 2.4.1, the fuel consumption of each type of engine in all months were calculated. By adding up the fuel consumption of all types of engines in each month, we can get the monthly fuel consumption and annual fuel consumption produced by domestic flights of civil aviation in China in 2010. The annual fuel consumption of 2010 is 12.12 million tons (metric tons). Fig. 1 shows the amount for each month.

Small fluctuation in fuel consumption (less than 7%) from the monthly mean (1.01 million tons) was observed for all months except Apr., partly arising from the variable monthly planning schedules, which are shifted by airlines monthly or seasonally to accommodate passenger demand. As can be seen from Fig. 1, the

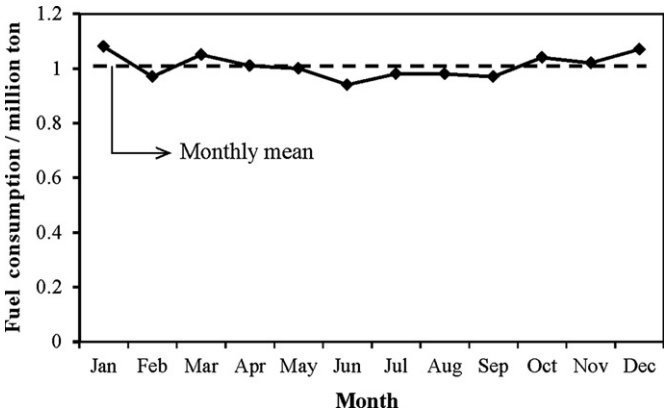


Fig. 1. Monthly fuel consumption in 2010 on domestic flights in China.

fuel consumption of Feb. is about 10% less than that of Jan. and Mar. This is because there are three days less in Feb than in Jan. and Mar.

The monthly fuel consumption in summer (from Jun. to Sep.) appears to be relatively small compared to that in the other seasons. Besides the variable monthly planning schedules, seasonal meteorological parameters could be the reason for this phenomenon. Engines consume less fuel with the rise of ambient temperature and the descent of ambient pressure, which can be deduced from Eqs. (1)–(3) in Section 2.4.1. This effect works mainly during LTO processes, while it's negligible during cruise process (Friedl et al., 1997).

3.2. Pollutant emissions and their monthly distribution

The emissions of HC, CO, NO_x, CO_2, and SO_2 produced by domestic flights of civil aviation in China in 2010 are 4600 tons, 39,700 tons, 154,100 tons, 38.21 million tons and 9700 tons, respectively. Monthly emissions of each pollutant are illustrated in Fig. 2.

As pollutant emissions are positively correlated with the fuel consumption, the monthly variation of pollutant emissions resembles that of fuel consumption (Fig. 1).

3.3. Contribution to pollutant emissions from various types of engines

Based on the above calculation method combined with the flight schedules and the aircraft/engine combination information, we can estimate the annual emissions of each pollutant emitted by various types of engines. The results are illustrated in Fig. 3.

As can be seen in Fig. 3, the largest NO_x, CO_2, SO_2 emissions are produced by the CFM56-7B27 engines, which correspond to the B737-800 aircrafts in this paper. This is because the majority of China's aircrafts are B737-800s, which execute a large number of domestic flights and have the longest total flight duration. The largest HC and CO emissions are produced by the CFM56-5B6 engines, which correspond to the aircraft type A320. The NO_x, CO_2, SO_2 annual emissions by CFM56-5B6 rank the second. This is because A320s owned by Chinese airlines are also quite numerous, and have a long total flight duration. However, the combustion efficiency of CFM56-5B6 might be lower than that of CFM56-7B27 so that the former HC and CO emissions are larger than the latter ones.

3.4. Contribution to pollutant emissions from various airlines

The annual fuel consumption and pollutant emissions of each airline are listed in Table 2.

The top five airlines in annual fuel consumption and pollutant emissions are listed in Fig. 4 for comparison.

It can be seen from Fig. 4 that the fuel consumption and pollutant emissions from the top five airlines account for 76% and 72–77% of the total because these airlines have a large amount of domestic flight missions and long total flight duration. China Southern Airline ranks first in fuel consumption and pollutant emissions, accounting for 27% and 25–28% of the total, respectively. Results presented in Fig. 4 also show that the rankings of emissions of different pollutants are not always the same. This is related to the configuration of fleet. The combustion efficiencies of each type of engine are different, so the emission indices for a particular pollutant are not the same. Airlines configuring engines of which combustion efficiencies are lower will emit more HC and CO. The top five airlines contribute to 73% and 72% emissions of HC and CO, respectively, obviously less than their contributions to other pollutants. This indicates that the combustion efficiencies of engines configured by these airlines are relatively high. Therefore, improving the configuration of fleet is a valid way of reducing pollutant emissions.

3.5. Uncertainty analysis

There are two main sources of uncertainties related to our estimation, uncertainties from flight schedules and uncertainties from simplifying assumptions. Although discrepancies between

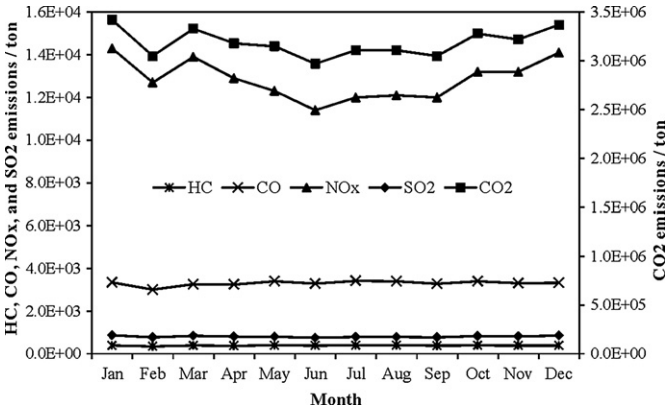


Fig. 2. Monthly emissions of HC, CO, NO_x, CO_2, and SO_2 in 2010 from domestic flights in China.

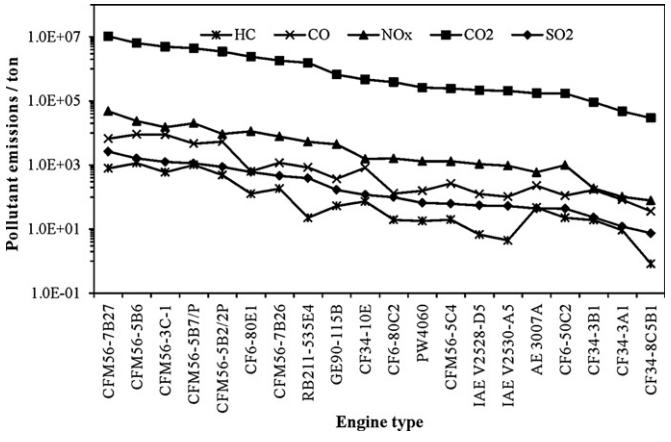


Fig. 3. Annual emissions of HC, CO, NO_x, CO_2 and SO_2 from various types of engines.

Table 2
Annual fuel consumption and pollutant emissions of each airline in China in 2010.

Airlines	Fuel consumption/ton	Annual emissions of each pollutant/ton				
		HC	CO	NO_x	CO_2	SO_2
China Southern	3,400,000	1290	10,100	45,500	10,700,000	2720
China Eastern	1,960,000	738	7430	23,900	6,170,000	1570
Air China	1,950,000	646	5360	25,700	6,140,000	1560
Shenzhen Airlines	1,270,000	456	3500	16,600	4,020,000	1020
Hainan Airlines	929,000	269	2640	12,500	2,930,000	743
Sichuan Airlines	497,000	277	1930	6110	1,570,000	398
Shandong Airlines	482,000	130	1380	5940	1,520,000	386
Xiamen Airlines	388,000	177	2420	4020	1,230,000	311
Shanghai Airlines	307,000	73	807	3950	968,000	246
Juneyao Airlines	161,000	87	656	1890	507,000	129
Beijing Capital Airlines	135,000	87	495	1820	427,000	108
Hong Kong Express	135,000	22	173	1890	427,000	108
Tianjin Airlines	129,000	72	651	1380	408,000	103
Dragonair	112,000	32	255	1480	353,000	90
Chengdu Airlines	83,300	57	280	1220	263,000	67
Lucky Air	82,700	41	444	888	261,000	66
China United Airlines	82,000	27	272	1110	259,000	66
Cathay Pacific Airways	74,100	23	201	985	234,000	59
Okay Airways	73,500	18	157	1060	232,000	59
Hebei Airlines	54,000	15	151	621	170,000	43
West China	46,400	26	157	616	146,000	37
Joy Air	45,300	27	177	590	143,000	36
Kunming Airlines	42,100	17	125	566	133,000	34
Air Macau	28,100	13	98	320	88,600	23
Grand China Air	28,000	7	59	394	88,200	22
Chongqing Airlines	26,200	15	89	343	82,600	21
Hong Kong Airlines	26,100	6	47	373	82,200	21
China Express Airlines	14,600	10	84	95	46,000	12
Spring Airlines	8970	4	34	109	28,300	7

actual operating schedules and planning schedules exist due to the influence factors such as weather, congestion, air traffic delays, etc., they have little impact on the total flight duration and distance, thus have little impact on the estimation results. On the other hand, simplifying assumptions in different aspects may have different influences on the estimation results. For instance, flight routes between city pairs are regarded as Great Circle routes and this may underestimate fuel consumption and pollutant emissions by a substantial amount for short-haul flights (Kettunen and Knorr, 2005). Winds are neglected in this paper, but its impact is relatively small (Friedl et al., 1997).

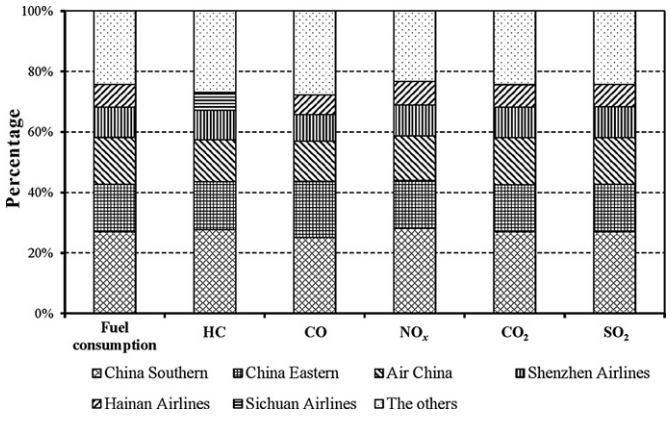


Fig. 4. Contribution to fuel consumption and pollutant emissions from airlines.

In order to verify the accuracy of our estimation results, we also calculated the fuel consumption in 2010 using the fuel consumption data in 2009 from the Development Research Center of the State Council (SCDRC) and the passenger turnover data from CAAC. The fuel consumption of China's civil aviation in 2009 is 13.14 million tons (SCDRC, 2011). In 2010, the passenger turnover of domestic flights accounts for 81% of the total passenger turnover, which is 16% more than that in 2009 (CAAC, 2011). Assuming that fuel consumption and passenger turnover are proportional, the fuel consumption from domestic flights of civil aviation in China in 2010 is $13.14 \times 116\% \times 81\% = 12.35$ million tons, from which the estimation result of 12.12 million tons has a 2% difference.

The calculation method used in this paper is BFFM2, which has been revised and improved by many researchers and international organizations. The method has been proved to be feasible. Therefore, the calculations in emission inventories, inaccurate as they may be, are of great value.

4. Conclusions

1) According to our estimation, the fuel consumption on domestic flights of civil aviation in China (excluding Taiwan Province) in 2010 is 12.12 million tons. Emissions of HC, CO, NO_x, CO_2, and SO_2 are 4600 tons, 39,700 tons, 154,100 tons, 38.21 million tons and 9700 tons respectively.
2) The B737-800s account for the largest amount of fuel consumption and NO_x, CO_2, and SO_2 emissions due to their numerous quantity owned by Chinese airlines and long flight duration.
3) The top five fuel consumers and CO, NO_x, CO_2, SO_2 emission producers are China Southern, Air China, China Eastern, Shenzhen Airlines and Hainan Airlines. The top five airlines in HC emissions are China Southern, China Eastern, Air China, Shenzhen Airlines and Sichuan Airlines. Fuel consumption and pollutant emissions are closely related to quantity of flight missions, total flight duration and fleet configuration. Changing for engines of high-combustion-efficiency is an efficient way to reduce pollutant emissions.

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